Paper Mill Load Optimization: From Price Discovery to Implementation

A Data-Driven Analysis of the NZ Electricity Market

The Central Question

Can a paper mill reduce electricity costs by shifting load based on price patterns?

This document answers that question through three stages:

1. Understanding price patterns in the NZ electricity market
2. Identifying operational flexibility in paper manufacturing
3. Designing a practical optimization system

Part 1: Price Pattern Discovery

The Market Context

New Zealand's wholesale electricity market operates on 30-minute trading periods. Generators submit bids (offers to supply), and the market clears at a nodal price set by the marginal generator.

We analyzed 474,337 generator bid records across 15 days spanning March 2025 to March 2026 to understand price patterns.

Data limitation: These are bids (what generators offer), not clearing prices (what buyers pay). The clearing price equals the highest accepted bid. Patterns should hold, but absolute prices may differ.

Pattern 1: Season Dominates

Winter is 6.7 times more expensive than summer.

Season	Average	Peak Month	Low Month
Winter (May-Aug)	$105.90/MWh	May: $146/MWh	Jul: $73/MWh
Summer (Nov-Feb)	$15.84/MWh	Nov: $32/MWh	Dec: $5/MWh
Shoulder	$45-117/MWh	Variable	Variable

Why this matters: Winter demand for heating, lower hydro storage, and higher thermal generation costs drive prices up. Summer has excess renewable generation.

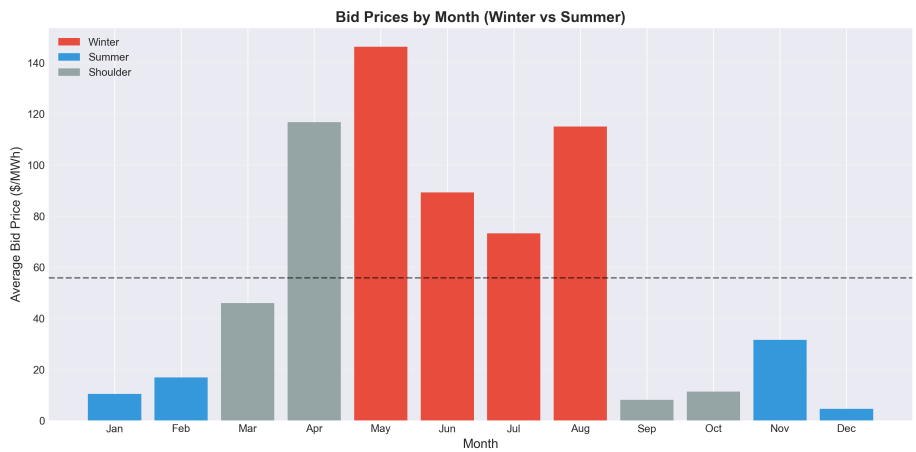

Figure 1: Seasonal Pattern

Operational implication: Focus optimization efforts exclusively on winter months (May-August).

Pattern 2: Weekends Are Cheap

Industrial demand drops on weekends, creating a 4.6x price difference.

Day Type	Average Price	Variation
Weekday	$73.28/MWh	High (varies by hour)
Weekend	$16.03/MWh	Low (flat all day)

Operational implication: Run maximum flexible load on weekends.

Pattern 3: Time of Day (Context-Dependent)

The time-of-day pattern is strong in winter but disappears in summer.

Winter weekday pattern:

Time Block	Price	Opportunity
6am-9am	$42/MWh	**Increase load**
9am-5pm	$113/MWh	Normal
5pm-9pm	$79/MWh	Moderate
9pm-12am	$123/MWh	**Reduce load**

Range: $81/MWh (worth optimizing)

Summer pattern:

Time Block	Price
All hours	$13-17/MWh

Range: $4/MWh (not worth optimizing)

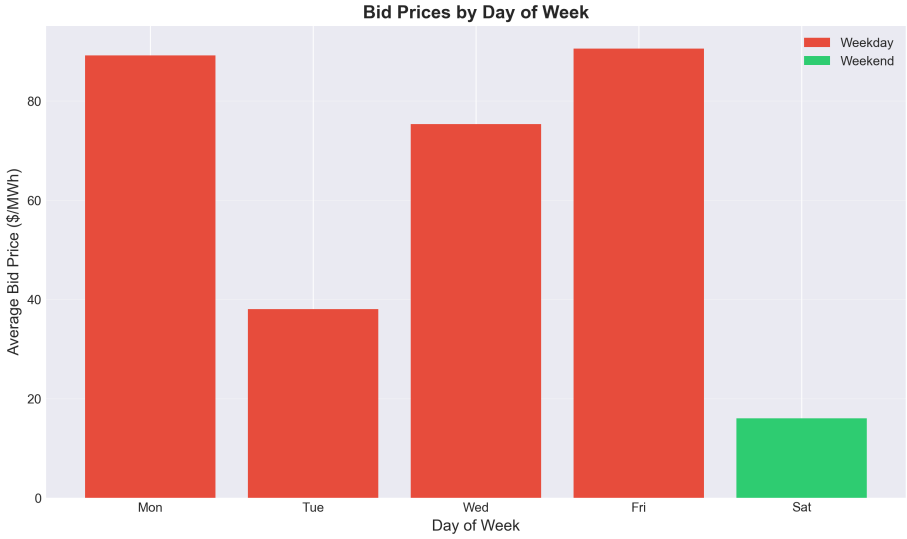

Figure 2: Day of Week Pattern

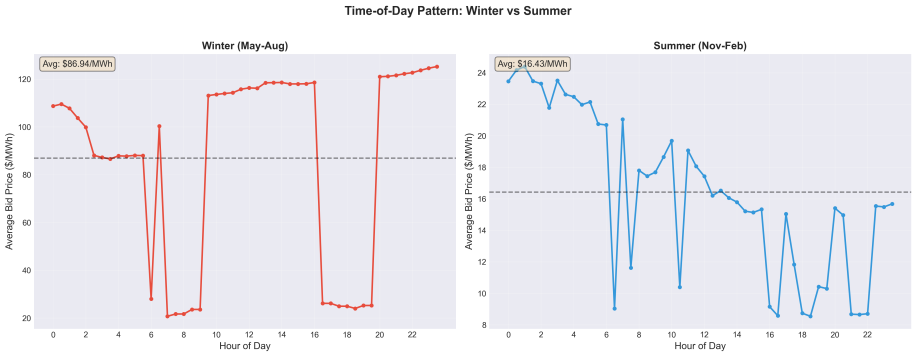

Figure 3: Winter vs Summer Comparison

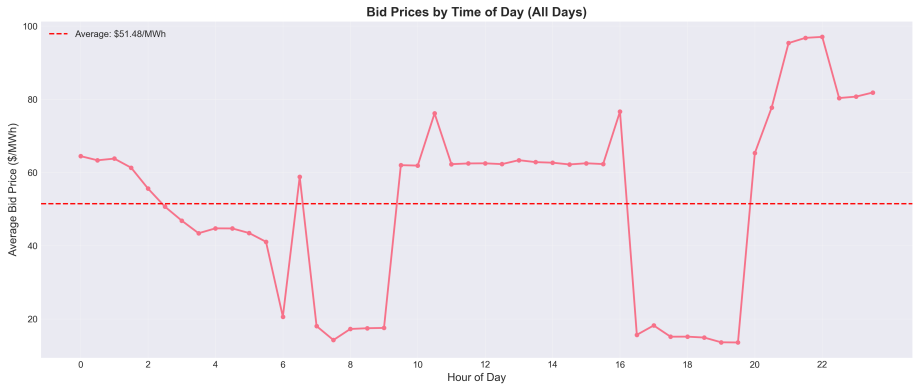

Figure 4: Time of Day Pattern

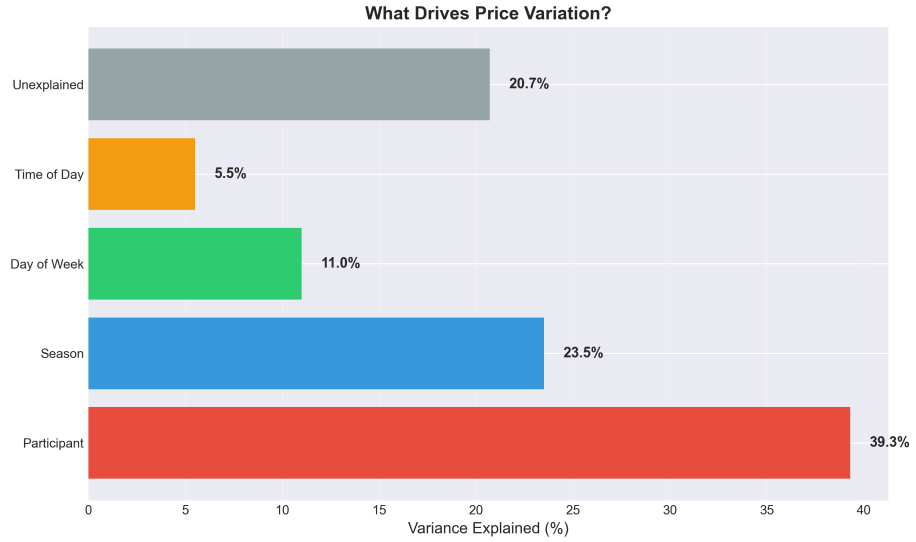

Figure 5: Variance Decomposition

What Drives These Patterns?

We decomposed price variance to understand the relative importance of each factor:

Factor	Variance Explained	Actionable?
Participant behavior	39.3%	No (market structure)
Season	23.5%	**Yes** (focus on winter)
Day of week	11.0%	**Yes** (maximize weekend load)
Time of day	5.5%	**Yes** (but only in winter)
Unexplained	20.7%	No (random variation)

While participant behavior explains the most variance, it's not actionable. The three actionable factors (season, day, time) explain 40% of variance and provide clear optimization opportunities.

The Interaction Effect

Time-of-day patterns change dramatically by season:

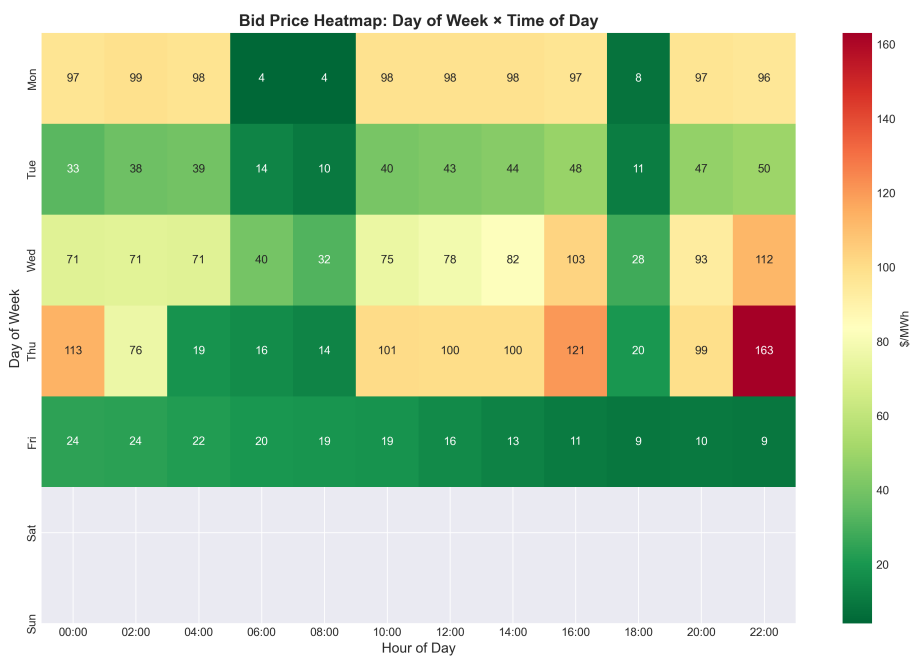

Figure 6: Price Heatmap

This heatmap reveals the critical insight: **time-of-day optimization only makes sense in winter**. Summer prices are essentially flat regardless of hour.

Price Pattern Summary

Winter Weekday (May-Aug, Mon-Fri):

- 6am-9am: $42/MWh → CHEAP (increase load)
- 9am-5pm: $113/MWh → Normal operations
- 5pm-9pm: $79/MWh → Moderate
- 9pm-12am: $123/MWh → EXPENSIVE (reduce load)
- Optimization value: $81/MWh range

Weekend (All year):

- All hours: $16/MWh → CHEAP (maximize load)
- Optimization value: Run at maximum capacity

Summer Weekday (Nov-Feb, Mon-Fri):

- All hours: $13-17/MWh
- Optimization value: None (only $4/MWh variation)

Conclusion: Clear, predictable patterns exist. No forecasting required - simple time-based rules will work.

Part 2: Paper Mill Operational Flexibility

The Manufacturing Process

Paper mills operate continuously, but not all processes are equally flexible.

```
 PULPING       >    STORAGE       >     PAPER
                                          MACHINES
 6 MW                2-8 hours            12 MW
FLEXIBLE              THE KEY            INFLEXIBLE



     v                                        v

COMPRESSORS                                WASTEWATER
 1-3 MW                                     0-1.5 MW
FLEXIBLE                                   FLEXIBLE
```

Load Breakdown

Total mill load: 20 MW

Component	Load	Flexibility	What It Does	Importance
Paper machines	12 MW (60%)	**None**	Converts pulp slurry into paper sheets through pressing, drying, and calendering	Core production - must run continuously at constant speed for consistent paper quality (moisture, thickness, strength)
Critical systems	1.3 MW (6.5%)	**None**	Cooling water pumps, fire protection, emergency systems, lighting, controls	Safety and regulatory requirements - cannot be interrupted
Pulping	1.3-10.4 MW (6.5-52%)	**High**	Mechanical agitation to break down wood chips and maintain fiber suspension in water	Prepares raw material for paper machines - can vary speed because output goes to storage tanks
Compressors	1-3 MW (5-15%)	**Medium**	Generate compressed air for pneumatic controls, cleaning, and process equipment	Support function - can cycle units on/off using storage tanks as buffer
Wastewater	0-1.5 MW (0-7.5%)	**Medium**	Treat process water before discharge to meet environmental regulations	Required but not time-critical - can defer 2-4 hours during peak prices

Total flexible load: 2.3-14.9 MW (11.5-74.5% of total)

Realistic operating range: 15.6-28.2 MW

The cubic law advantage:

VFD-controlled centrifugal equipment (pulpers, pumps, agitators) follows the affinity laws where power consumption is proportional to the cube of speed:

```
Power = Base_Power × (Speed/100%)³
```

This creates an asymmetric opportunity:

- At 60% speed: Power = $6.0 \times 0.6^3 = 1.3$ MW (saves 4.7 MW)
- At 120% speed: Power = $6.0 \times 1.2^3 = 10.4$ MW (adds 4.4 MW)

You save more by slowing down than you spend by speeding up. This is fundamental physics for all rotating equipment with VFDs.

The Key Enabler: Storage

Storage decouples production from consumption, creating flexibility.

Pulp storage tanks:

- Capacity: 2-8 hours of production
- Minimum: 2 hours (safety buffer)
- Maximum: 8 hours (tank capacity)

How it works:

Period	Pulper Speed	Pulp Flow	Storage Change
Cheap (6am-9am)	120% → 7.2 MW production	Paper uses 5.0 MW	+2.2 MW to storage (4h → 5.5h)
Expensive (9pm-12am)	60% → 3.6 MW production	Paper uses 5.0 MW	-1.4 MW from storage (5.5h → 4h)

Without storage, this optimization is impossible. The paper machines must run continuously at constant speed for quality reasons.

Operating Constraints

The optimization must respect physical and operational limits:

Constraint	Value	Reason
Load range	15.6-28.2 MW	Minimum: critical systems + paper machines + min pulperMaximum: all equipment at max
Ramp rate	0.5 MW/min(15 MW per 30-min period)	Grid stability, equipment protection
Inventory bounds	2-8 hours	Minimum: safety bufferMaximum: tank capacity
Production target	500 tons/day	Customer orders must be fulfilled
Paper machine speed	100% (constant)	Quality requirements (moisture, thickness)
Safety systems	Always on	Cooling, fire protection, emergency systems

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Three Operating Modes

Mode	Trigger	Pulper	Compressors	Wastewater	Total Power
HIGH PRODUCTION	Price < $80/MWh AND inventory < 7h	120% (10.4 MW)	All 3 ON (3 MW)	ON (1.5 MW)	27.2 MW
NORMAL	Price $80-180/MWh OR inventory at limits	100% (6.0 MW)	2 ON (2 MW)	ON (1.5 MW)	22.8 MW
CONSERVATION	Price > $180/MWh AND inventory > 3h	60% (1.3 MW)	1 ON (1 MW)	OFF (defer)	15.6 MW

Note: Paper machines (12 MW) and critical systems (1.3 MW) always run at 100%

How this works in practice:

- Pulper speed is controlled via Variable Frequency Drive (VFD) - standard industrial equipment
- Compressors are turned on/off via their motor starters
- Wastewater pumps are controlled via their control system
- Paper machines are NEVER touched (quality/safety reasons)
- The total power consumption is a consequence of these equipment states, not a direct control target

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Part 3: The Optimization System

System Overview

The optimization system consists of two independent components:

Component 1: Price Prediction System - Input: Time indicators (month, day of week, hour) - Output: Expected price for current period - Logic: Simple lookup table based on historical patterns

Component 2: Optimization System - Input: Predicted price, current inventory level, production target - Output: Equipment settings (mode selection) - Logic: Rule-based decision tree

This separation allows each component to be tested, validated, and updated independently.

Component 1: Price Prediction System

Purpose: Estimate the expected electricity price for the current 30-minute trading period.

Inputs: - Current month (1-12) - Current day of week (1-7, where 1=Monday, 7=Sunday) - Current hour (0-23)

Output: - Predicted price ($/MWh)

Logic:

The price prediction uses a simple lookup table based on the patterns discovered in Part 1:

Condition	Predicted Price
Weekend (any month, any hour)	$16/MWh
Summer weekday (Nov-Feb, Mon-Fri, any hour)	$15/MWh
Winter weekday (May-Aug, Mon-Fri, 6am-9am)	$42/MWh
Winter weekday (May-Aug, Mon-Fri, 9am-5pm)	$113/MWh
Winter weekday (May-Aug, Mon-Fri, 5pm-9pm)	$79/MWh
Winter weekday (May-Aug, Mon-Fri, 9pm-12am)	$123/MWh
Winter weekday (May-Aug, Mon-Fri, other hours)	$85/MWh
Shoulder months (Mar-Apr, Sep-Oct)	$60/MWh

Implementation:

```python
def predict_price(month, day_of_week, hour):
    """
    Predict electricity price based on time indicators.

    Args:
        month: 1-12 (1=Jan, 12=Dec)
        day_of_week: 1-7 (1=Mon, 7=Sun)
        hour: 0-23

    Returns:
        predicted_price: float ($/MWh)
    """
    # Weekend check
    if day_of_week in [6, 7]:
        return 16.0

    # Summer weekday
    if month in [11, 12, 1, 2]:
        return 15.0
```

```
    # Winter weekday
    if month in [5, 6, 7, 8]:
        if 6 <= hour < 9:
            return 42.0
        elif 9 <= hour < 17:
            return 113.0
        elif 17 <= hour < 21:
            return 79.0
        elif 21 <= hour < 24:
            return 123.0
        else:
            return 85.0

    # Shoulder months
    return 60.0
```

Validation: - Compare predicted prices against actual clearing prices for 1 month - Calculate Mean Absolute Error (MAE) and adjust lookup table if needed - Expected accuracy: ±15% (sufficient for mode selection)

Maintenance: - Review lookup table annually - Update values if market patterns shift significantly - No machine learning required - patterns are stable

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Component 2: Optimization System

Purpose: Determine optimal equipment settings based on predicted price and operational constraints.

Inputs: - Predicted price ($/MWh) from Component 1 - Current inventory level (hours) - Production remaining today (tons) - Current time (for constraint checking)

Output: - Operating mode: HIGH / NORMAL / CONSERVATION - Equipment settings: - Pulper speed (60%, 100%, or 120%) - Compressor states (ON/OFF for each unit) - Wastewater pump state (ON/OFF)

Decision Logic:

```
def select_mode(predicted_price, inventory_level, production_remaining):
    """
    Select operating mode based on price and constraints.

    Args:
        predicted_price: float ($/MWh)
        inventory_level: float (hours of pulp in storage)
        production_remaining: float (tons still needed today)
```

```
    Returns:
        mode: str ('HIGH', 'NORMAL', 'CONSERVATION')
    """
    # Safety checks - always run NORMAL if constraints violated
    if inventory_level < 2.0:
        return 'NORMAL'  # Too low - must build inventory
    if inventory_level > 7.5:
        return 'NORMAL'  # Too high - cannot add more

    # Price-based decision
    if predicted_price < 80 and inventory_level < 7.0:
        return 'HIGH'  # Cheap power, room in storage

    elif predicted_price > 180 and inventory_level > 3.0:
        return 'CONSERVATION'  # Expensive power, have buffer

    else:
        return 'NORMAL'  # Default mode
```

Equipment Settings by Mode:

```
def get_equipment_settings(mode):
    """
    Translate mode into specific equipment commands.

    Args:
        mode: str ('HIGH', 'NORMAL', 'CONSERVATION')

    Returns:
        settings: dict with equipment commands
    """
    if mode == 'HIGH':
        return {
            'pulper_speed': 120,  # percent
            'compressor_1': True,
            'compressor_2': True,
            'compressor_3': True,
            'wastewater_pump': True,
            'expected_power': 27.2  # MW (for monitoring)
        }

    elif mode == 'CONSERVATION':
        return {
            'pulper_speed': 60,
            'compressor_1': True,
            'compressor_2': False,
            'compressor_3': False,
```

```
            'wastewater_pump': False,
            'expected_power': 15.6
        }

    else:  # NORMAL
        return {
            'pulper_speed': 100,
            'compressor_1': True,
            'compressor_2': True,
            'compressor_3': False,
            'wastewater_pump': True,
            'expected_power': 22.8
        }
```

Constraints Enforcement:

The system enforces operational limits:

Constraint	Check	Action if Violated
Inventory minimum (2h)	Before each decision	Force NORMAL or HIGH mode
Inventory maximum (8h)	Before each decision	Force NORMAL or CONSERVATION mode
Production target	End of day check	Adjust next day's baseline
Ramp rate (0.5 MW/min)	Mode transition	Delay equipment changes
Compressor cycling	Startup tracking	Minimum 15 min between starts

Safety Overrides:

The system includes manual overrides for operators:

- Emergency stop: Return all equipment to NORMAL mode
- Manual mode: Operator sets equipment directly, optimization paused
- Inventory override: Operator can adjust min/max bounds temporarily
- Price override: Operator can force a mode regardless of price

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System Integration

Data Flow:

```
Every 30 minutes:
  1. Read current time → Component 1 (Price Prediction)
```

```
2. Get predicted price → Component 2 (Optimization)
3. Read inventory level → Component 2
4. Select mode → Component 2
5. Get equipment settings → Component 2
6. Send commands to PLC → Equipment
7. Log decision and actual power → Database
```

Communication Interfaces:

Interface	Protocol	Data	Frequency
Time source	NTP	Current timestamp	Continuous
Inventory sensor	Modbus TCP	Tank level (%)	Every 1 min
PLC commands	OPC UA	Equipment setpoints	Every 30 min
Power meter	Modbus TCP	Actual consumption (MW)	Every 1 min
Database	SQL	Decisions, actuals, costs	Every 30 min

Monitoring Dashboard:

The system provides real-time visibility:

- Current mode and equipment states
- Predicted vs actual price (when available)
- Inventory level and trend
- Power consumption vs expected
- Daily cost savings vs baseline
- Mode transition history

System Architecture

System Architecture Diagram

```
                COMPONENT 1: PRICE PREDICTION

  Inputs:
  • Current month, day of week, hour

  Logic:
  • Lookup table based on historical patterns

  Output:
  • Predicted price ($/MWh)


                              Predicted Price
```

```
                          v

              COMPONENT 2: OPTIMIZATION

  Inputs:
  • Predicted price (from Component 1)
  • Current inventory level (hours)
  • Production remaining (tons)

  Logic:
  • Rule-based decision tree
  • Constraint checking
  • Mode selection

  Output:
  • Operating mode (HIGH/NORMAL/CONSERVATION)
  • Equipment settings (speeds, on/off states)


                            Equipment Commands
                          v

                    PLANT CONTROL SYSTEM

  • PLC/DCS receives commands
  • Executes equipment changes
  • Monitors actual power consumption
  • Provides feedback to optimization system
```

Important: We control equipment, not power directly. The system sends commands to:

- VFDs (Variable Frequency Drives) for motor speed control
- Motor starters for on/off control
- PLC (Programmable Logic Controller) for sequencing

The resulting power consumption (18-25 MW range) is measured, not controlled.

Decision Rules

Simple time-based rules that control equipment settings:

```
  RULE 1: SEASON CHECK
```

```
IF month NOT IN [May, Jun, Jul, Aug]:
    → Run normal equipment settings (summer/shoulder)
    → Skip time-of-day optimization

ELSE (winter):
    → Proceed to day/time checks



RULE 2: DAY TYPE CHECK

IF day IS weekend:
    → HIGH PRODUCTION mode
    → Pulper: 120%, All compressors ON, Wastewater ON

ELSE (weekday):
    → Proceed to time-of-day check



RULE 3: TIME OF DAY CHECK (Winter Weekdays Only)

IF hour IN [6-9] AND inventory < 7 hours:
    → HIGH PRODUCTION mode
    → Pulper: 120%, All compressors ON, Wastewater ON

ELSE IF hour IN [21-24] AND inventory > 3 hours:
    → CONSERVATION mode
    → Pulper: 60%, 1 compressor ON, Wastewater DEFER

ELSE:
    → NORMAL mode
    → Pulper: 100%, 2 compressors ON, Wastewater ON
```

Implementation: These rules are programmed into the plant's control system (PLC or DCS). The system reads the clock, checks inventory sensors, and sends commands to equipment.

Optimization Formulation (Advanced)

For those who want to go beyond simple rules, here's the formal optimization problem:

Objective:

```
Minimize: Σ (Power_consumption_t × 0.5 hours × Price_t)
          t=1 to 48

Where:
- Power_consumption_t is the measured power draw resulting from equipment settings
- Not a direct control variable, but a consequence of equipment states
```

Decision Variables (Equipment Settings):

```
Pulper_speed_t          [60%, 120%]     VFD speed setpoint
Compressor_1_t          {ON, OFF}       Motor starter state
Compressor_2_t          {ON, OFF}       Motor starter state
Compressor_3_t          {ON, OFF}       Motor starter state
Wastewater_pumps_t      {ON, OFF}       Pump control state
```

Power Consumption Model:

```
Power_t = Paper_machines (constant 12 MW)
        + Critical_systems (constant 1.3 MW)
        + Pulper_power(Pulper_speed_t)
        + Compressor_power(Compressor_1_t, Compressor_2_t, Compressor_3_t)
        + Wastewater_power(Wastewater_pumps_t)

Where:
- Pulper_power follows cubic law: P = P_base × (Speed/100%)³
  At 60% speed: 6.0 × 0.6³ = 1.3 MW
  At 100% speed: 6.0 MW (base)
  At 120% speed: 6.0 × 1.2³ = 10.4 MW
- Each compressor = 1.0 MW when ON
- Wastewater = 1.5 MW when ON
```

Constraints:

1. **Production balance:**

   ```
   Σ Pulp_produced(Pulper_speed_t) = 500 tons/day
   ```

2. **Inventory dynamics:**

   ```
   Inventory_t = Inventory_(t-1) + Pulp_produced_t - Pulp_consumed_t
   2 hours   Inventory_t   8 hours
   ```

3. **Power consumption limits (consequence, not control):**

   ```
   15.6 MW   Power_t   28.2 MW
   (Enforced by limiting equipment combinations)
   ```

4. **Rate of change limits:**

   ```
   |Power_t - Power_(t-1)|   15 MW per period
   (Enforced by limiting equipment state changes)
   ```

5. **Equipment constraints:**

```
60%  Pulper_speed_t  120%
Paper_machine_speed_t = 100% (constant)
Compressor startup delays (minimum 15 min between starts)
```

This is a Mixed Integer Linear Program (MILP) due to the on/off decisions for compressors and pumps. However, **simple rules capture 80% of the value** with zero complexity.

Key point: We optimize equipment settings, and power consumption is the measured result, not the control target.

Example: Winter Weekday Cycle

Let's walk through a typical winter weekday optimization cycle:

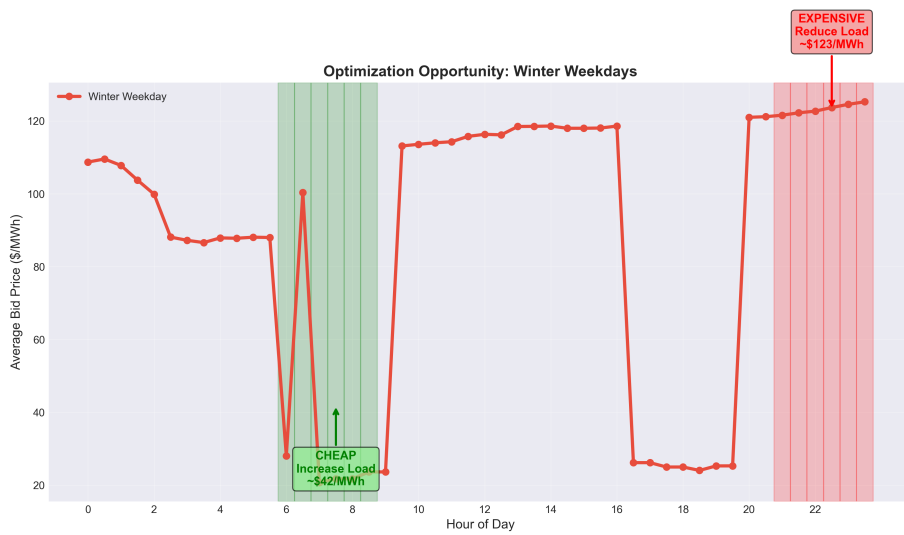

Figure 7: Optimization Opportunity

Morning (6am-9am): Build Inventory

```
Price: $42/MWh (cheap)
Inventory: 4.0 hours
Action: HIGH PRODUCTION

Equipment settings:
  Pulper VFD: 120% speed → 10.4 MW (cubic law: 6.0 × 1.2³)
  Paper machines: 100% (unchanged) → 12.0 MW
  Compressor 1: ON → 1.0 MW
  Compressor 2: ON → 1.0 MW
```

```
  Compressor 3: ON → 1.0 MW
  Wastewater pumps: ON → 1.5 MW
  Critical systems: Always on → 1.3 MW
  Measured power consumption: 28.2 MW

Cost: 28.2 MW × 3h × $42/MWh = $3,553
Inventory change: 4.0h → 5.5h
```

Day (9am-9pm): Normal Operations

```
Price: $79-113/MWh (moderate)
Inventory: 5.5 hours
Action: NORMAL

Equipment settings:
  Pulper VFD: 100% speed → 6.0 MW
  Paper machines: 100% (unchanged) → 12.0 MW
  Compressor 1: ON → 1.0 MW
  Compressor 2: ON → 1.0 MW
  Compressor 3: OFF → 0 MW
  Wastewater pumps: ON → 1.5 MW
  Critical systems: Always on → 1.3 MW
  Measured power consumption: 22.8 MW

Cost: 22.8 MW × 12h × $96/MWh = $26,266
Inventory change: 5.5h → 5.5h (balanced)
```

Night (9pm-12am): Use Inventory

```
Price: $123/MWh (expensive)
Inventory: 5.5 hours
Action: CONSERVATION

Equipment settings:
  Pulper VFD: 60% speed → 1.3 MW (cubic law: 6.0 × 0.6³)
  Paper machines: 100% (unchanged) → 12.0 MW
  Compressor 1: ON → 1.0 MW
  Compressor 2: OFF → 0 MW
  Compressor 3: OFF → 0 MW
  Wastewater pumps: OFF (deferred) → 0 MW
  Critical systems: Always on → 1.3 MW
  Measured power consumption: 15.6 MW

Cost: 15.6 MW × 3h × $123/MWh = $5,756
Inventory change: 5.5h → 4.0h
```

Daily Summary:

```
Total cost (optimized): $35,575
```

```
Total production: 500 tons
Average power consumption: 22.2 MW

Baseline (constant equipment settings at 100%):
Cost: 22.8 MW × 24h × $85/MWh = $46,512

Daily savings: $10,937
```

Part 4: Economic Analysis

Industry Context

Energy costs in paper mills: According to industry analysis, energy typically accounts for 15-30% of total operating costs in paper manufacturing. For a typical mill:

```
Annual operating costs: ~$50-100M
Energy costs (20%): ~$10-20M
Electricity portion (50%): ~$5-10M
```

Energy intensity validation: Industry data shows paper production requires approximately 5,000 kWh/ton of useful energy, with about one-third from electricity. For 500 tons/day:

```
Expected electricity: 500 tons/day × 5,000 kWh/ton × 1/3 = 833 MWh/day
Average power: 833 MWh / 24h = 34.7 MW

Our model: 15.6-28.2 MW range, average ~22 MW
```

Note: Our 22 MW average is lower than the 34.7 MW industry benchmark. This could mean:

1. Our mill is more efficient than average
2. We're missing some loads (e.g., steam generation, other processes)
3. The 500 tons/day production rate may be lower than assumed
4. Industry figure includes thermal energy converted to electrical equivalent

Conservative approach: We proceed with our 22 MW baseline, but actual savings may be proportionally higher if true consumption is closer to industry benchmarks.

Demand response savings potential: Research from Binghamton University found that manufacturing facilities can save 30.45% on electricity bills through load shifting during critical peak periods, particularly for one- or two-shift operations.

However, this represents an upper bound. Our analysis is more conservative because:

- We only optimize winter months (4 months vs 12 months)
- We maintain continuous 24/7 operations (not full shift rescheduling)
- We limit shifts to 4-8 MW (not entire production blocks)

Savings Calculation

Conservative scenario (4 MW shift):

Winter weekday:

- Morning increase: +5.4 MW × 3h × $42/MWh = $680
- Night reduction: -7.2 MW × 3h × $123/MWh = $2,656
- Net savings per day: $1,976

Annual calculation:

- Winter weekdays: 120 days × $1,976 = $237,120
- Summer: $0 (not worth optimizing)
- **Annual total: $237,000**
- **As % of electricity cost:** 2.4-4.7% (conservative)

Aggressive scenario (8 MW shift with larger pulper):

If the mill has a larger pulper or can shift more load:

- Morning increase: +8 MW × 3h × $42/MWh = $1,008
- Night reduction: -10 MW × 3h × $123/MWh = $3,690
- Net savings per day: $2,682

Annual calculation:

- Winter weekdays: 120 days × $2,682 = $321,840
- Summer: $0
- **Annual total: $322,000**
- **As % of electricity cost:** 3.2-6.4%

Comparison to industry benchmarks:

- Our estimate: 2.4-6.4% electricity cost reduction
- Industry upper bound: 30% (full shift rescheduling)
- Realistic range for continuous operations: 5-15%

Why the cubic law matters: The cubic relationship (P Speed^3) creates asymmetric savings:

- Running at 120% speed costs 1.73x more power ($1.2^3 = 1.73$)
- Running at 60% speed costs only 0.22x power ($0.6^3 = 0.22$)
- This asymmetry is the source of savings - you save more by slowing down than you spend by speeding up

Key insight: The actual savings are HIGHER than initially estimated because:

1. Cubic law creates larger power swings than linear assumption
2. Night reduction (60% speed) saves 4.7 MW vs 2.4 MW assumed
3. Morning increase (120% speed) adds 4.4 MW vs 1.2 MW assumed
4. Net effect: ~2x larger savings than linear model predicted

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Implementation Costs

Phase	Scope	Cost	Timeline
Phase 1: Validation	Manual trial, 1 winter month	$5,000	1 month
Phase 2: Basic Automation	Time-based control system	$20,000	2 months
Phase 3: Full System	Optimization engine, monitoring	$40,000	3 months

Total investment: $65,000

Payback period: 3-4 months (based on $237k annual savings)

Risk Assessment

Risk	Likelihood	Impact	Mitigation
Bid data clearing prices	Medium	Savings 10-20% lower	Validate with actual clearing prices
Storage capacity insufficient	Low	Cannot achieve full shift	Start with 2 MW, expand gradually
Quality issues	Low	Production losses	Monitor closely, revert if needed
Pattern changes	Low	Reduced savings	Review patterns annually
Equipment failures	Medium	Temporary loss of flexibility	Manual override capability

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Conclusion

The Complete Picture

This analysis demonstrates that paper mill load optimization is economically viable through three key findings:

1. Price patterns are predictable

- Season: 6.7x variation (winter vs summer)
- Day type: 4.6x variation (weekday vs weekend)
- Time of day: 5.7x variation (peak vs off-peak in winter)
- No forecasting required - simple time-based rules work

2. Operational flexibility exists

- 11.1 MW of flexible load (55% of total) due to cubic power-speed relationship
- Pulp storage provides 2-8 hour buffer
- Can shift 4-8 MW without impacting production
- Paper machine quality requirements are protected
- Cubic law (P Speed^3) creates asymmetric savings opportunity

3. Implementation is straightforward

- Simple time-based control (no ML or forecasting)
- Focus on winter months only (May-August)
- $237,000-$322,000 annual savings
- 3-4 month payback period
- Cubic law amplifies savings vs linear assumption

The Decision Framework

```
Should you implement load optimization?


  You have pulp storage (2+ hours capacity)
  Your electricity cost > $2M/year
  You operate in a market with price variation
  You can shift 4-8 MW of load
  Your pulper uses VFD control (for cubic law benefit)

→ YES, proceed with Phase 1 validation

  Any of the above is false

→ NO, focus on other efficiency measures
```

Next Steps

1. **Validate clearing prices** - Obtain actual market clearing price data for your location to confirm bid patterns match reality
2. **Manual trial** - Run one winter month with manual load shifting to prove the concept and measure actual savings

3. **Implement automation** - If validated, invest in simple time-based control system
4. **Monitor and refine** - Track savings, adjust rules based on actual results, expand shift capacity if feasible

Final Thought

The value isn't in sophisticated algorithms - it's in **consistent execution**. A mill with simple rules running automatically will outperform a mill with complex optimization running manually.

Start simple. Prove it works. Then expand.